Net change in money

1 Describe what it means if the net change in money is:

a Positive

b Negative

2 For each of the following, describe the change in money:

a After buying and then selling a bicycle, the net change in Laura's money was $-\$44$.

b After buying and then selling a scarf, the net change in Sandy's money was $\$5$.

c After buying and then selling a chair, the net change in Derek's money was $-\$8$.

d After buying and then selling a scarf, the net change in Bianca's money was $\$0$.

e After buying and then selling a toaster, the net change in Edward's money was $\$65$.

3 For each of the following items, find the net change in money as a directed number:

a Valentina bought a pillow for $\$2$ and sold it for $\$16$.

b Bill bought a chair for $\$17$ and sold it for $\$6$.

c Hannah bought a tennis racquet for $\$82$ and sold it for $\$169$.

d Aoife bought a car for $\$12\,000$ and sold it five years later for $\$4500$.

e Yan bought an apartment for $\$380\,000$ and sold it ten years later for $\$520\,000$.

4 In its opening month, a local pizza store sold 190 large pizzas at $\$20$ each and 200 small pizzas at $\$6$ each. The store incurred electricity costs of $\$26$, water costs of $\$10$, rent of $\$408$, wages of $\$1345$ and ingredient costs of $\$210$.

a Calculate the total revenue of the store in the first month.

b Calculate the total expenses of the store in the first month.

c Calculate the net change in money by the store in the first month.

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Cost and sale price

5 Find the sale price of the following items:

- **a** Cost price: $\$58$, profit: $\$10$.
- **b** Cost price: $\$120$, profit: $\$32.50$.
- **c** Cost price: $\$80$, loss: $\$27$.
- **d** Cost price: $\$250$, loss: $\$135$.
- **e** Sandy bought a stove for $\$238$. She eventually sold the stove and made a profit of $\$39$.
- **f** William bought a wallet for $\$12$. He eventually sold the wallet and made a profit of $\$2$.
- **g** Pauline bought a plane ticket for $\$469$. She then had to sell the plane ticket and made a loss of $\$9$.
- **h** Ryan bought a guitar for $\$176$. He eventually sold the guitar and made a loss of $\$37$.

6 Find the cost price of the following items:

- **a** Sale price: $\$72$, profit: $\$15$.
- **b** Sale price: $\$10\,000$, profit: $\$2200$.
- **c** Sale price: $\$520$, loss: $\$210$.
- **d** Sale price: $\$780$, loss: $\$360$.
- **e** Ben sold a toothbrush for $\$12$ and made a profit of $\$3$.
- **f** Pauline sold a washing machine for $\$486$ and made a loss of $\$7$.
- **g** Sarah sold a laptop for $\$233$ and made a profit of $\$55$.
- **h** David sold a pair of earrings for $\$161$ and made a loss of $\$55$.

7 Find the profit on the following items:

- **a** Cost price: $\$65$, sale price: $\$82$.
- **b** Cost price: $\$120$, sale price: $\$186$.
- **c** Cost price: $\$1400$, sale price: $\$2050$.
- **d** Cost price: $\$86.50$, sale price: $\$93.50$.
- **e** Glen bought a trading card for $\$11$ and sold it for $\$24$.
- **f** Roxanne bought a dress for $\$712$ and sold it for $\$829$.

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8 Find the loss on the following items:

a Cost price: $\$70$, sale price: $\$45$.

b Cost price: $\$225$, sale price: $\$138$.

c Cost price: $\$5870$, sale price: $\$4250$.

d Cost price: $\$850$, sale price: $\$375.50$.

e Judy bought a basketball for $\$99$ and sold it for $\$92$.

f Charlie bought a sculpture for $\$660$ and sold it for $\$585$.

9 Find the profit (or loss) for each of the following:

a The sale price is $\$540$ and the cost price is $\$284$.

b The sale price is $\$2065$ and the cost price is $\$3120$.

c The sale price is $\$340.50$ and the cost price is $\$290.80$.

d The sale price is $\$1800$ and the cost price is $\$3500$.

e The cost price is $\$276.90$ and the sale price is $\$320.55$.

f The cost price is $\$2480$ and the sale price is $\$1770$.

Percentage profit and loss

10 Charlie bought a cake for $\$220$ and sold it for $\$209$. Describe Charlie's change in money after buying and selling the cake.

11 Sean bought a spade for $\$36$ and sold it for $\$45$. Describe Sean's change in money after buying and selling the spade.

12 Oliver bought a baseball bat for $\$130$ and sold it for $\$143$.

a Calculate Oliver's profit.

b Find the percentage profit of the cost price of the baseball bat.

c Describe Oliver's change in money after buying and selling the baseball bat.

13 Bianca bought a microwave for $\$50$ and sold it for $\$35$.

a Calculate Bianca's loss.

b Find the percentage loss of the cost price of the microwave.

c Describe Bianca's change in money after buying and selling the microwave.

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14 John buys a used laptop for $\$391$ and resells it for $\$554$.

- **a** Calculate the profit John make on the laptop.
- **b** Find this profit as a percentage of the cost price.

15 Paul invests $\$9000$ in the stock market only to see his shares plummet in value to $\$4509$.

- **a** Calculate the loss Paul make on the shares.
- **b** Find this loss as a percentage of the cost price.

16 Sally purchased a dress for $\$20$ and auctioned it off on eBay for $\$41$. Find:

- **a** The profit (or loss).
- **b** The profit (or loss) as a percentage of the cost, correct to two decimal places.
- **c** The profit (or loss) as a percentage of the sale price to two decimal places.

17 Buzz bought a mobile phone for $\$170$ and sold it 7 months later, making a loss of 30% on the purchase price. Find the selling price mobile phone.

18 Every item in store is sold at 8% above cost. If a product was bought from the manufacturer for $\$360$, find:

- **a** The sale price
- **b** The profit (or loss)

19 Sarah bought a bicycle for $\$400$ and sold it for a 18% profit.

- **a** Find the profit made on the bicycle.
- **b** Hence, find the sale price of the bicycle.

20 Hermione bought a phone for $\$200$ and sold it for a 16% loss.

- **a** Find the loss made on the phone.
- **b** Hence, find the sale price of the phone.

21 Noah sold a jumping castle for $\$700$ which was a 12% profit.

- **a** Find the sale price of the jumping castle as a percentage of the cost price.
- **b** Hence, find the cost price of the jumping castle.

22 James sold a jacket for $\$600$ which was a 4% loss.

- **a** Find the sale price of the jacket as a percentage of the cost price.
- **b** Hence, find the cost price of the jacket.

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23 Yvonne bought a car for $\$17\,600$ and sold it 2 years later, making a loss of 19%. Find the sale price of the car.

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